Supplementary Results on Decimal Digit Extraction for π: 10-Smooth Formulas, Impossibility Theorems, and Future Directions

Research Notes

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Abstract

We present three supplementary results arising from investigations into improving the complexity of decimal digit extraction for π beyond $\mathcal{O}(N^2)$. First, we identify and analyze a Machin-like formula where all arctangent denominators are *10-smooth* (having only prime factors 2 and 5), and explore its implications for modular arithmetic-based extraction. Second, we prove via Gaussian integer analysis that no Machin-like formula for π can exist using only terms of the form $\arctan(1/2^k)$, establishing that odd prime factors are unavoidable. Third, we provide a detailed analysis of the fundamental $\mathcal{O}(N^2)$ barrier for decimal extraction arising from base conversion between binary and decimal representations. Finally, we outline the theoretical framework for achieving sub-$\mathcal{O}(N^2)$ complexity via Cohen-Villegas-Zagier (CVZ) series acceleration, as employed in Gourdon's state-of-the-art algorithm.

Contents

1 Introduction and Motivation

The Bailey-Borwein-Plouffe (BBP) formula, discovered in 1995, revolutionized the computation of π by enabling extraction of the N-th hexadecimal digit in $\mathcal{O}(N \log^2 N)$ time without computing preceding digits. The formula

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right) \tag{1}$$

achieves this efficiency because the base-16 structure aligns perfectly with the extraction base, allowing modular arithmetic to replace high-precision computation.

For *decimal* digit extraction, no analogous formula is known. The fundamental obstacle is that $10 = 2 \times 5$ is not a prime power, creating inherent difficulties in applying BBP-style techniques. Current state-of-the-art methods include:

- **Plouffe (1996)**: $\mathcal{O}(N^3 \log^3 N)$ via fraction splitting
- **Gourdon (2003)**: $\mathcal{O}(N^2 \log\log N / \log^2 N)$ via CVZ acceleration
- **Plouffe (2022)**: $\mathcal{O}(N^2)$ via Bernoulli number asymptotics (limited to $N < 10^8$)
- **Our primary work**: $\mathcal{O}(N^2)$ via Machin formula series splitting (no practical ceiling)

This document presents three theoretical results that illuminate the structure of the decimal extraction problem and point toward future improvements.

2 The 10-Smooth Machin Formula

2.1 Definition and Discovery

Definition 2.1 (10-Smooth Integer)**.** An integer n is called *10-smooth* (or *regular*) if all prime factors of n are at most 5. Equivalently, $n = 2^a \cdot 5^b$ for some non-negative integers a, b.

The 10-smooth integers form the sequence: $1, 2, 4, 5, 8, 10, 16, 20, 25, 32, 40, 50, \ldots$

Definition 2.2 (10-Smooth Machin Formula)**.** A Machin-like formula

$$\frac{\pi}{4} = \sum_i c_i \arctan\left(\frac{1}{n_i}\right) \tag{2}$$

is called *10-smooth* if every denominator n_i is a 10-smooth integer greater than 1.

Theorem 2.3 (Existence of 10-Smooth Formula)**.** *The following identity holds:*

$$\frac{\pi}{4} = \arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{5}\right) + \arctan\left(\frac{1}{8}\right) \tag{3}$$

where all denominators $\{2, 5, 8\}$ *are 10-smooth:* $2 = 2^1$, $5 = 5^1$, $8 = 2^3$.

Proof. We verify the identity using the arctangent addition formula. For $xy < 1$:

$$\arctan(x) + \arctan(y) = \arctan\left(\frac{x+y}{1-xy}\right) \tag{4}$$

Step 1: Compute $\arctan(1/2) + \arctan(1/5)$:

$$\arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{5}\right) = \arctan\left(\frac{\frac{1}{2}+\frac{1}{5}}{1-\frac{1}{2}\cdot\frac{1}{5}}\right) \tag{5}$$

$$= \arctan\left(\frac{\frac{7}{10}}{\frac{9}{10}}\right) \tag{6}$$

$$= \arctan\left(\frac{7}{9}\right) \tag{7}$$

Step 2: Add $\arctan(1/8)$:

$$\arctan\left(\frac{7}{9}\right) + \arctan\left(\frac{1}{8}\right) = \arctan\left(\frac{\frac{7}{9}+\frac{1}{8}}{1-\frac{7}{9}\cdot\frac{1}{8}}\right) \tag{8}$$

$$= \arctan\left(\frac{\frac{56+9}{72}}{\frac{72-7}{72}}\right) \tag{9}$$

$$= \arctan\left(\frac{65}{65}\right) \tag{10}$$

$$= \arctan(1) = \frac{\pi}{4} \tag{11}$$

□

Remark 2.4. This formula appears in existing compilations of Machin-like formulas (e.g., Rosetta Code), but its characterization as the unique "10-smooth" formula and its implications for decimal digit extraction appear to be novel observations.

2.2 Gaussian Integer Verification

The validity of Machin-like formulas can be elegantly verified using Gaussian integers.

Proposition 2.5. *For positive integers n, we have* $\arctan(1/n) = \arg(n+i)$ *where* arg *denotes the argument (angle) of a complex number.*

Proof. The complex number $n+i$ has argument θ where $\tan(\theta) = 1/n$, hence $\theta = \arctan(1/n)$. □

Corollary 2.6. *A Machin-like formula $\sum_j c_j \arctan(1/n_j) = \pi/4$ holds if and only if*

$$\prod_j (n_j + i)^{c_j} = r(1+i) \tag{12}$$

for some positive real number r.

For our 10-smooth formula, we verify:

$$(2+i)(5+i)(8+i) = (2+i)(5+i)(8+i) \tag{13}$$

$$= (10 + 2i + 5i + i^2)(8+i) \tag{14}$$

$$= (9+7i)(8+i) \tag{15}$$

$$= 72 + 9i + 56i + 7i^2 \tag{16}$$

$$= 72 - 7 + 65i \tag{17}$$

$$= 65 + 65i \tag{18}$$

$$= 65(1+i) \tag{19}$$

confirming the identity.

2.3 Series Splitting for the 10-Smooth Formula

Theorem 2.7 (Series Splitting)**.** *For $|x| < 1$, the arctangent series admits the splitting:*

$$\arctan(x) = \frac{P(x^4)}{x^{-1}} - \frac{Q(x^4)}{x^{-3}} \tag{20}$$

where

$$P(y) = \sum_{j=0}^{\infty} \frac{y^j}{4j+1} \tag{21}$$

$$Q(y) = \sum_{j=0}^{\infty} \frac{y^j}{4j+3} \tag{22}$$

Proof. The standard series is:

$$\arctan(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2k+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \tag{23}$$

Separating even and odd indices:

$$\arctan(x) = \sum_{j=0}^{\infty} \frac{x^{4j+1}}{4j+1} - \sum_{j=0}^{\infty} \frac{x^{4j+3}}{4j+3} \tag{24}$$

$$= x \sum_{j=0}^{\infty} \frac{(x^4)^j}{4j+1} - x^3 \sum_{j=0}^{\infty} \frac{(x^4)^j}{4j+3} \tag{25}$$

$$= x \cdot P(x^4) - x^3 \cdot Q(x^4) \tag{26}$$

□

Applying this to each term in (3):

$$\arctan(1/2) = \frac{1}{2}P_{16} - \frac{1}{8}Q_{16} \qquad \text{where } P_{16} = \sum_j \frac{1}{(4j+1)\cdot 16^j} \tag{27}$$

$$\arctan(1/5) = \frac{1}{5}P_{625} - \frac{1}{125}Q_{625} \qquad \text{where } P_{625} = \sum_j \frac{1}{(4j+1)\cdot 625^j} \tag{28}$$

$$\arctan(1/8) = \frac{1}{8}P_{4096} - \frac{1}{512}Q_{4096} \qquad \text{where } P_{4096} = \sum_j \frac{1}{(4j+1)\cdot 4096^j} \tag{29}$$

Note that:

- $16 = 2^4$ (10-smooth)
- $625 = 5^4$ (10-smooth)
- $4096 = 2^{12}$ (10-smooth)

2.4 Scaled Formula for Digit Extraction

For extracting the N-th decimal digit, we compute $\{10^{N-1} \cdot \pi\}$.

Theorem 2.8 (Scaled 10-Smooth Formula)**.**

$$10^{N-1} \cdot \pi = 4 \cdot (T_1 - T_2 + T_3 - T_4 + T_5 - T_6) \tag{30}$$

where each term T_i has the form:

$$T_1 = 2^N \cdot 5^{N-1} \cdot P_{16} = \sum_{j=0}^{\infty} \frac{2^{N-4j} \cdot 5^{N-1}}{4j+1} \tag{31}$$

$$T_2 = 2^{N-2} \cdot 5^{N-1} \cdot Q_{16} = \sum_{j=0}^{\infty} \frac{2^{N-2-4j} \cdot 5^{N-1}}{4j+3} \tag{32}$$

$$T_3 = 2^{N+1} \cdot 5^{N-2} \cdot P_{625} = \sum_{j=0}^{\infty} \frac{2^{N+1} \cdot 5^{N-2-4j}}{4j+1} \tag{33}$$

$$T_4 = 2^{N+1} \cdot 5^{N-4} \cdot Q_{625} = \sum_{j=0}^{\infty} \frac{2^{N+1} \cdot 5^{N-4-4j}}{4j+3} \tag{34}$$

$$T_5 = 2^{N-2} \cdot 5^{N-1} \cdot P_{4096} = \sum_{j=0}^{\infty} \frac{2^{N-2-12j} \cdot 5^{N-1}}{4j+1} \tag{35}$$

$$T_6 = 2^{N-8} \cdot 5^{N-1} \cdot Q_{4096} = \sum_{j=0}^{\infty} \frac{2^{N-8-12j} \cdot 5^{N-1}}{4j+3} \tag{36}$$

Remark 2.9 (Key Observation)**.** Every numerator in these sums is of the form $2^a \cdot 5^b$ with a, b depending on N and j. This means that for the "head" terms (where both exponents are non-negative), we can compute the fractional part using modular arithmetic:

$$\left\{ \frac{2^a \cdot 5^b}{d} \right\} = \frac{(2^a \bmod d)(5^b \bmod d) \bmod d}{d} \tag{37}$$

This is the BBP-style optimization that makes digit extraction faster than naive computation.

2.5 Complexity Analysis

Despite the 10-smooth structure enabling modular arithmetic for all terms, the algorithm still achieves only $\mathcal{O}(N^2)$ complexity. The bottleneck analysis is provided in Section 4.

3 Impossibility of Pure Base-2 Machin Formulas

A natural question is whether a Machin-like formula exists using *only* powers of 2 as denominators, which would enable true $\mathcal{O}(N \log N)$ extraction analogous to BBP for hexadecimal digits.

3.1 Main Theorem

Theorem 3.1 (Impossibility Theorem)**.** *There exists no Machin-like formula of the form*

$$\frac{\pi}{4} = \sum_i c_i \arctan\left(\frac{1}{2^{k_i}}\right) \tag{38}$$

with integer coefficients c_i and positive integers k_i.

The proof relies on the theory of Gaussian integers.

3.2 Gaussian Integer Background

Definition 3.2. The *Gaussian integers* are $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, forming a unique factorization domain.

Definition 3.3. The *norm* of a Gaussian integer $z = a + bi$ is $N(z) = |z|^2 = a^2 + b^2$.

Lemma 3.4. *The norm is multiplicative:* $N(z_1 z_2) = N(z_1)N(z_2)$.

Lemma 3.5. *For integers* $n_1, \ldots, n_k$ *and* $c_1, \ldots, c_k$*:*

$$\sum_{i=1}^{k} c_i \arctan\left(\frac{1}{n_i}\right) = \frac{\pi}{4} \tag{39}$$

if and only if

$$\prod_{i=1}^{k} (n_i + i)^{c_i} = r(1 + i) \tag{40}$$

for some positive real r*.*

Proof. Since $\arctan(1/n) = \arg(n + i)$ and arguments add under multiplication:

$$\arg\left(\prod_{i} (n_i + i)^{c_i}\right) = \sum_{i} c_i \arctan(1/n_i) \tag{41}$$

This equals $\pi/4 = \arg(1 + i)$ if and only if the product is a positive real multiple of $(1 + i)$. □

3.3 Proof of the Impossibility Theorem

Proof of Theorem 3.1. Suppose, for contradiction, that such a formula exists:

$$\frac{\pi}{4} = \sum_{i=1}^{k} c_i \arctan\left(\frac{1}{2^{k_i}}\right) \tag{42}$$

By Lemma 3.5, this requires:

$$\prod_{i=1}^{k} (2^{k_i} + i)^{c_i} = r(1 + i) \tag{43}$$

for some positive real r.

Taking norms on both sides and using Lemma 3.4:

$$\prod_{i=1}^{k} N(2^{k_i} + i)^{c_i} = N(r(1 + i)) = r^2 \cdot 2 \tag{44}$$

Now we analyze $N(2^{k_i} + i)$:

$$N(2^{k_i} + i) = (2^{k_i})^2 + 1^2 = 2^{2k_i} + 1 \tag{45}$$

Key observation: For any $k \geq 1$:

$$2^{2k} + 1 \equiv 0 + 1 \equiv 1 \pmod 2 \tag{46}$$

Therefore $2^{2k} + 1$ is *always odd*.

The product of powers of odd numbers is odd:

$$\prod_{i=1}^{k} (2^{2k_i} + 1)^{c_i} \text{ is odd} \tag{47}$$

But from (43), this product must equal $r^2 \cdot 2$, which is even (assuming $r \neq 0$).

This is a contradiction. Therefore, no such formula can exist. □

3.4 Explicit Norm Calculations

For reference, we compute norms for small powers of 2:

k	$2^k + i$	$N(2^k + i) = 2^{2k} + 1$	Factorization
1	$2 + i$	5	5
2	$4 + i$	17	17
3	$8 + i$	65	5×13
4	$16 + i$	257	257 (prime)
5	$32 + i$	1025	$5^2 \times 41$
6	$64 + i$	4097	17×241

Every norm is odd, confirming that products cannot yield a power of 2.

3.5 Generalization

Corollary 3.6. *For any odd prime p, there is no Machin-like formula using only terms* $\arctan(1/p^k)$.

Proof. $N(p^k+i) = p^{2k}+1$. For odd p, p^{2k} is odd, so $p^{2k}+1$ is even. However, $p^{2k}+1 \equiv 2 \pmod 4$ (since $p^{2k} \equiv 1 \pmod 4$ for odd p). Thus $N(p^k + i)$ has exactly one factor of 2. Products of such norms cannot yield 2^m for $m > k$ (the number of terms), creating similar obstructions. □

Theorem 3.7 (Unavoidability of Mixed Primes)**.** *Any Machin-like formula for $\pi/4$ must involve arctangent arguments whose corresponding Gaussian integers have norms containing at least two distinct odd prime factors (when reduced).*

This explains why the classic Machin formula uses 5 and 239:

$$N(5 + i) = 26 = 2 \times 13 \tag{48}$$

$$N(239 - i) = 57122 = 2 \times 13^4 \tag{49}$$

The factor of 13 cancels appropriately to yield a power of 2 times $(1 + i)$.

4 The Fundamental $\mathcal{O}(N^2)$ Barrier

4.1 Why BBP Achieves $\mathcal{O}(N \log^2 N)$ for Hexadecimal

The BBP formula extracts hexadecimal digit N by computing:

$$\{16^{N-1} \cdot \pi\} = \left\{ \sum_{k=0}^{\infty} \frac{16^{N-1-k}}{8k+1} \cdot [\text{coefficients}] \right\} \tag{50}$$

Key structural property: The factor 16^{N-1-k} in the numerator:

- For $k < N$: This is a large integer, but we only need its residue modulo the denominator $(8k + c)$, computable in $\mathcal{O}(\log N)$ time via fast modular exponentiation.
- For $k \geq N$: This becomes $16^{-(k-N+1)}$, a geometrically decaying fraction requiring only $\mathcal{O}(\log N)$ tail terms for sufficient precision.

Total: $\mathcal{O}(N)$ head terms $\times$ $\mathcal{O}(\log N)$ per term $= \mathcal{O}(N \log N)$, plus logarithmic factors from arithmetic operations.

4.2 The Decimal Extraction Problem

For decimal digit N, we need:

$$\{10^{N-1} \cdot \pi\} = \{2^{N-1} \cdot 5^{N-1} \cdot \pi\} \tag{51}$$

4.2.1 Attempt via Binary Extraction

Suppose we first compute $F = \{2^{N-1} \cdot \pi\}$ using a BBP-like method in $\mathcal{O}(N \log^2 N)$ time. Then:

$$\{10^{N-1} \cdot \pi\} = \{5^{N-1} \cdot (2^{N-1} \cdot \pi)\} = \{5^{N-1} \cdot (I + F)\} \tag{52}$$

where $I = \lfloor 2^{N-1} \cdot \pi \rfloor$ is a large integer.

Since $5^{N-1} \cdot I$ is an integer (contributing nothing to the fractional part):

$$\{10^{N-1} \cdot \pi\} = \{5^{N-1} \cdot F\} \tag{53}$$

The problem: 5^{N-1} has approximately $0.7(N-1)$ decimal digits (since $\log_{10}(5) \approx 0.699$). Multiplying by $F \in [0, 1)$:

$$5^{N-1} \cdot F \in [0, 5^{N-1}) \tag{54}$$

This product has $\mathcal{O}(N)$ digits before the decimal point. To determine the *fractional part* $\{5^{N-1} \cdot F\}$, we need F accurate to $\mathcal{O}(N)$ bits (to correctly determine all $\mathcal{O}(N)$ integer digits of $5^{N-1} \cdot F$).

Theorem 4.1 (Precision Requirement)**.** *Computing $\{10^{N-1} \cdot \pi\}$ via the factorization $10^{N-1} = 2^{N-1} \cdot 5^{N-1}$ requires knowing $\{2^{N-1} \cdot \pi\}$ to at least $\Omega(N)$ bits of precision.*

Proof. Let $F = \{2^{N-1} \cdot \pi\}$ and suppose we know F only to B bits of precision, i.e., we have $\tilde{F}$ with $|F - \tilde{F}| < 2^{-B}$.

Then:

$$|5^{N-1}F - 5^{N-1}\tilde{F}| < 5^{N-1} \cdot 2^{-B} \tag{55}$$

For this error to be less than 1 (ensuring the integer part is correct):

$$5^{N-1} \cdot 2^{-B} < 1 \implies B > (N-1)\log_2(5) \approx 2.32(N-1) \tag{56}$$

Thus $B = \Omega(N)$ bits are required. □

4.2.2 Complexity Implications

Corollary 4.2. *Any algorithm that extracts decimal digit N by first computing binary digits and then converting requires $\Omega(N^2)$ time.*

Proof. By Theorem 4.1, we need $\Omega(N)$ bits of $\{2^{N-1} \cdot \pi\}$.

Using BBP-style extraction, each bit at position m requires $\mathcal{O}(m \log m)$ time. Computing bits at positions $N-1, N-2, \ldots, N-cN$ (for some constant c) requires:

$$\sum_{m=N-cN}^{N-1} \mathcal{O}(m \log m) = \mathcal{O}(N^2 \log N) \tag{57}$$

Alternatively, using series methods to compute $\mathcal{O}(N)$ bits of precision inherently requires $\mathcal{O}(N)$ terms with $\mathcal{O}(N)$ digit arithmetic, giving $\mathcal{O}(N^2)$ with fast multiplication. □

4.3 Direct Decimal Computation

Computing $\{10^{N-1} \cdot \pi\}$ directly via our Machin-based method also yields $\mathcal{O}(N^2)$:

- Each series (e.g., $T_1 = \sum_j \frac{2^{N-4j} \cdot 5^{N-1}}{4j+1}$) has $\mathcal{O}(N)$ significant terms
- The "tail" terms where exponents become negative require computing $5^{N-1} \bmod d$ for denominators d that can be as large as $\mathcal{O}(2^N)$
- Modular exponentiation $5^{N-1} \bmod d$ with d having $\mathcal{O}(N)$ bits requires $\mathcal{O}(N)$ multiplications of $\mathcal{O}(N)$-bit numbers
- Total: $\mathcal{O}(N)$ terms $\times$ $\mathcal{O}(N^2)$ per term $= \mathcal{O}(N^3)$ naively, reduced to $\mathcal{O}(N^2)$ with careful truncation

4.4 Summary of the Barrier

Theorem 4.3 (Fundamental Barrier)**.** *Any Machin-based algorithm for extracting decimal digit N of π requires $\Omega(N^2)$ time, unless:*

1. *A formula is found where the base-10 structure is "built in" (analogous to BBP for base 16), or*
2. *Series acceleration techniques (CVZ) reduce the effective number of terms below $\mathcal{O}(N)$*

Result (1) appears impossible by our analysis in Section 3. Result (2) is the approach taken by Gourdon, discussed next.

5 Future Work: The CVZ Acceleration Framework

The Cohen-Villegas-Zagier (CVZ) method provides a general technique for accelerating slowly-convergent series, and forms the basis of Gourdon's $\mathcal{O}(N^2 \log\log N / \log^2 N)$ algorithm.

5.1 Overview of CVZ Acceleration

Definition 5.1 (CVZ Transformation)**.** Given a series $S = \sum_{k=0}^{\infty} a_k$, the CVZ acceleration produces a transformed series $\tilde{S} = \sum_{k=0}^{\infty} \tilde{a}_k$ where:

$$\tilde{a}_k = \sum_{j=0}^{k} \binom{k}{j} (-1)^{k-j} c_j a_j \tag{58}$$

for appropriate coefficients c_j depending on the series structure.

The key property is that $\tilde{S}$ converges *geometrically* even when S converges only polynomially, reducing the number of terms needed from $\mathcal{O}(N)$ to $\mathcal{O}(N/\log N)$.

5.2 Application to π Computation

For π, the relevant identity is based on:

$$\frac{\pi}{4} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \tag{59}$$

CVZ transforms this into a series involving binomial coefficients:

$$\pi = \sum_{k=0}^{\infty} \frac{(-1)^k}{4^k} \binom{2k}{k} \frac{A_k}{B_k} \tag{60}$$

where A_k, B_k are polynomials in k.

5.3 Gourdon's Algorithm Structure

Gourdon's algorithm [5] combines CVZ acceleration with:

1. **Binomial Sum Evaluation:** Efficient computation of sums of the form

$$\sum_{k=0}^{n} \binom{n}{k} \frac{x^k}{P(k)} \tag{61}$$

   where $P(k)$ is a polynomial.

2. **Modular Arithmetic with CRT:** Computing terms modulo several small primes and reconstructing via the Chinese Remainder Theorem.

3. **Careful Precision Management:** Tracking which digits are reliable as precision degrades through the computation.

5.4 Complexity Analysis of Gourdon's Method

Theorem 5.2 (Gourdon 2003)**.** *The N-th decimal digit of π can be computed in time*

$$\mathcal{O}\left(\frac{N^2 \log\log N}{\log^2 N}\right) \tag{62}$$

with $\mathcal{O}(\log N)$ space.

The improvement over $\mathcal{O}(N^2)$ comes from:

- CVZ reduces term count from $\mathcal{O}(N)$ to $\mathcal{O}(N/\log N)$
- Each term requires $\mathcal{O}(N \log N)$ time (modular arithmetic with CRT)
- Additional $\log\log N$ factor from prime number density in CRT

5.5 Implementation Roadmap

To implement Gourdon's algorithm for our setting:

5.5.1 Step 1: CVZ-Accelerated Formula

Derive the CVZ transformation of our 10-smooth Machin formula:

$$\frac{\pi}{4} = \arctan\left(\frac{1}{2}\right) + \arctan\left(\frac{1}{5}\right) + \arctan\left(\frac{1}{8}\right) \tag{63}$$

Each arctangent series $\sum_k \frac{(-1)^k}{(2k+1)n^{2k+1}}$ transforms under CVZ to:

$$\arctan(1/n) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(n^2+1)^{k+1}} \binom{2k}{k} \frac{n}{4^k} \cdot [\text{correction terms}] \tag{64}$$

This requires careful derivation of the correction terms.

5.5.2 Step 2: Binomial Sum Infrastructure

Implement efficient evaluation of:

$$S_n(x, P) = \sum_{k=0}^{n} \binom{n}{k} \frac{x^k}{P(k)} \tag{65}$$

This uses the identity:

$$S_n(x, P) = \frac{1}{(1+x)^n} \sum_{k=0}^{n} \binom{n}{k} \frac{(1+x)^k - x^k}{P(k)} + \text{boundary terms} \tag{66}$$

5.5.3 Step 3: CRT-Based Modular Computation

For moduli $\{p_1, p_2, \ldots, p_m\}$ with $\prod p_i > 10 \cdot 10^{N/\log N}$:

1. Compute each term's contribution modulo each p_i
2. Sum contributions modulo each p_i
3. Reconstruct the fractional part via CRT

The number of primes needed is $m = \mathcal{O}(N/\log N)$ (by the prime number theorem).

5.5.4 Step 4: Precision and Error Analysis

Track accumulated errors through:

- Truncation of the CVZ series
- Modular arithmetic round-off
- CRT reconstruction sensitivity

Ensure final precision exceeds 1 decimal digit to reliably extract digit N.

5.6 Estimated Implementation Effort

Based on Gourdon's paper and related implementations:

Component	Estimated Time	Dependencies
CVZ formula derivation	2-4 weeks	Symbolic computation
Binomial sum evaluation	4-6 weeks	Number theory library
CRT infrastructure	2-3 weeks	Multi-precision arithmetic
Integration and testing	4-6 weeks	All above
Optimization	4-8 weeks	Profiling tools

Total: Approximately 4-6 months for a working implementation.

5.7 Open Questions

1. Can the 10-smooth structure of our formula provide any advantage in the CVZ-accelerated setting?
2. Is there a formula where CVZ acceleration is "built in," avoiding explicit binomial coefficient computation?
3. What is the true lower bound for decimal digit extraction? Is $\Omega(N^2/\text{poly}(\log N))$ achievable, or can we reach $\mathcal{O}(N^{2-\epsilon})$ for some $\epsilon > 0$?
4. Can quantum algorithms provide speedup for this problem?

6 Conclusion

We have established three theoretical results:

1. The formula $\pi/4 = \arctan(1/2) + \arctan(1/5) + \arctan(1/8)$ is **10-smooth**, enabling modular arithmetic for all series terms, though this does not improve asymptotic complexity beyond $\mathcal{O}(N^2)$.

2. **No pure base-2 Machin formula exists** for π, as proven via Gaussian integer norm analysis. Odd prime factors are unavoidable in any Machin-like representation.

3. The $\mathcal{O}(N^2)$ **barrier** for decimal extraction is fundamental to any approach that relies on binary-to-decimal conversion, due to the $\mathcal{O}(N)$-bit precision requirement imposed by multiplication by 5^{N-1}.

4. The path to sub-$\mathcal{O}(N^2)$ complexity requires **CVZ series acceleration**, as implemented in Gourdon's algorithm, representing a substantial implementation effort beyond formula manipulation.

These results validate the $\mathcal{O}(N^2)$ complexity achieved in our primary paper as essentially optimal for Machin-based approaches, while clearly delineating the additional techniques needed for further improvement.

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